

Predicting Stock Splits Using Ensemble Machine Learning and SMOTE Oversampling

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Abstract

This study predicts stock splits using two ensemble machine learning techniques: gradient boosting machines (GBMs) and random forests (RFs). The goal is to form implementable portfolios based on positive predictions to generate abnormal returns. Since splits are rare events, we use SMOTE oversampling to synthesize new observations of splits in the sample to improve predictions. When predicting stock splits in the next quarter, GBM and RF achieve area under the receiver operating characteristic curve (AUC) scores of around 0.86 and 0.87, respectively. GBM and RF predictions generate monthly five-factor alphas (Fama and French, 2015) of 0.26% and 0.95% among stocks in the smallest size quintile. Three important features for predicting stock splits in both ensemble ML methods are current price levels, the ratio of current price to the price at last split, and stock returns in the past twelve months. When predicting stock splits in the next year, GBMs generate monthly five-factor alphas of 0.38% among small stocks.

JEL classification: G11; G14; G17

Keywords: stock splits, ensemble machine learning, gradient boosting machines, random forests, SHAP feature importance, hyperparameter tuning, SMOTE oversampling

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1. Introduction

The finance literature documents significant abnormal stock returns around stock-split announcements.¹ For example, Fama et al. (1969) find an abnormal return of 34.07% in the 29 months before stock splits. Lakonishok and Lev (1987) find an excess return of 53% in the five years before stock splits.² Examples of other studies that find positive abnormal returns before, around, and after stock splits include Grinblatt, Masulis, and Titman (1984), Lamoureux and Poon (1987), Ikenberry, Rankine, and Stice (1996), and Li and Ye (2021), among others. Given these return patterns, we ask whether machine learning (ML) techniques can predict stock splits. If yes, we ask whether portfolios formed on the predictions of the ML techniques generate abnormal returns.

Using logit regressions as the benchmark, we test the following two ensemble ML methods: Gradient Boosting Machines (GBMs) and Random Forest (RFs). We study ML for a couple of reasons. First, ML classifiers do not impose a functional form between the dependent variable (i.e., the target, in this paper, whether a firm splits its stock) and the independent variables (the features). In contrast, traditional econometric classifiers such as probit or logit regressions assume a linear relationship between the target and the features. In particular, we choose ensemble MLs in this study because they train multiple learners and combine them into one powerful composite prediction model. Ensemble ML techniques usually make better predictions than a single learner. The common methods used in ensemble MLs are either *bagging* or *boosting*, with RFs belonging to the first

¹ From here onward, we will refer to stock-split announcements as stock splits since they are the events we are predicting.

² Numerous studies have confirmed strong stock performance in the months before stock splits (Asquith, Healey, and Palepu, 1989; McNichols and Dravid, 1990; Maloney and Mulherin, 1992).

category and GBMs the second.³ Second, since stock splits are rare events, ML methods such as oversampling can improve the predictions.

We use 82 quarters (1980Q1 to 2000Q2) of data to train the models, and 82 quarters (2000Q3 to 2020Q4) of data to make out-of-sample predictions. To evaluate the performance of each model, we use the AUC score. The AUC score is the area under the receiver operating characteristic (ROC) curve; it can be interpreted as the probability that the model will assign a higher stock-split likelihood to a stock that splits than to a stock that does not split. To evaluate the performance of portfolios formed on the predictions of the ML models, we use five-factor alphas (Fama and French, 2015). We also use CAPM, the three-factor model (Fama and French, 1993) and a six-factor model (by adding the momentum factor to the five-factor model) as robustness tests.

When predicting stock splits in the next quarter, both GBMs and RFs achieve AUC scores of 0.85 without oversampling. In contrast, the logit regression achieves an AUC score of only 0.75.⁴ Among the twelve features that we use to predict stock splits, three of them are important in both ML methods: the current price level of the stock, the ratio of the current stock price to the price at last split,⁵ and the stock return in the past twelve months. All three features are positively related to the probability of conducting a stock split.

³ The idea behind *bagging* is to aggregate predictions from individual learners to form a consensus, taking advantage of the “wisdom of the crowd.” In contrast, *boosting* uses an ensemble of weak prediction models to build a stronger model in a stage-wise manner. Specifically, in each stage, the method focuses on the weak prediction models in the previous stage and improves them. We will discuss in detail later how RFs and GBMs work when predicting stock splits.

⁴ If one were to make random predictions, the AUC score is about 0.5, whereas a perfect prediction results in an AUC score of 1. Hence an improvement of the AUC score from 0.75 to 0.85 is economically meaningful.

⁵ In case a stock has never split before, we use the first daily closing price available in CRSP.

However, despite the high AUC scores, the next-quarter predictions from neither GBMs nor RFs translate to a profitable trading strategy: portfolios formed on their next-quarter predictions do not generate abnormal returns.⁶ This is likely because the true positive predictions of the ML models are diluted by their false positive predictions. The unconditional probability of a split in the next quarter is only 1.82%. So even when we hold the false positive rate low while classifying stock splits using the ML models, the number of false positive predictions is too large in the portfolio to generate alphas.

Because stock splits are rare events, our sample is extremely imbalanced: there are many more negative events than positive events. We use the SMOTE oversampling technique to improve the performance of the prediction models.⁷ We use costed error functions to decide the optimal proportion of oversampling for each model. The oversampling method improves the performance of all three models: the AUC scores from Logit, GBM, and RF increase to 0.79, 0.86, and 0.87, respectively. GBM and RF predictions augmented by oversampling generate monthly five-factor alphas (Fama and French, 2015) of 0.26% and 0.95% among stocks in the smallest size quintile.

We then check the performance of the two ensemble ML methods in predicting stock splits in the *next year*. The AUC scores from both GBMs and RFs are around 0.86.

⁶ Specifically, at the end of each calendar quarter, we predict the probability of a stock-split announcement in the next quarter. We then buy stocks that our models classify as splitters and hold the stocks for a quarter. The monthly equal-weighted and value-weighted portfolio returns are then regressed on the Fama-French five factors. The abnormal returns are the intercept from the regressions.

⁷ In traditional random oversampling, duplicating examples of the rare class are added to the sample to increase the number of the minority class. A more effective oversampling method, SMOTE oversample, developed by Chawla et al. (2002), synthesizes examples of the rare class. Specifically, it first chooses an observation from the rare class and randomly selects an example from its k nearest neighbors. A synthetic example is then created as a convex combination of the two instances in the feature space. See Chawla et al. (2002) for more details. Studies have shown that SMOTE oversampling performs better than random oversampling (e.g., Fernandez et al., 2018; Ogundimu, 2021) and is considered the *de facto* method in learning from imbalanced data.

In contrast, the logit regression achieves an AUC score of 0.78. Portfolios formed on stock-split predictions *for the next year* using the GBM technique can generate abnormal returns, especially among smaller stocks. Further tests show that the abnormal returns are indeed due to GBMs' ability to successfully predict stock splits in the next year: true positive predictions generate high returns while false positive predictions don't. Investors can design implementable portfolios based on the above results: equal-weighted and value-weighted portfolios formed on GBM predictions generate monthly five-factor alphas of 0.38% and 0.33% (Fama and French, 2015), respectively, among stocks in the smallest size quintile. Neither RFs nor logit regressions can generate portfolios with abnormal returns when predicting stock splits in the next year. SMOTE oversampling doesn't help the performance of any of the three models when predicting splits in the next year. The unconditional probability of a split is 6.63% in our sample, and this makes splits in the next year not an extremely rare event. Our results suggest that oversampling helps only in extremely rare events.

Our results are robust to different specifications. If we use CAPM, the Fama-French three-factor model, or add the momentum factor to the Fama-French five-factor model, or if we use different time periods, results remain qualitatively the same.

To the best of our knowledge, this is the first paper to use ML methods to predict stock splits. Prior literature uses traditional econometric methods, such as probit or logit regressions, to predict stock splits (e.g., Baker, Greenwood, and Wurgler, 2009; Perez, Shkilko, and Tang, 2011). By using ML methods to predict stock splits, we join other researchers who are applying ML to finance (e.g., Erel et al., 2021; Li et al., 2021; see Section 2 for more examples). The results in this paper show that ML methods are

promising in improving the performance of predicting corporate events compared to traditional econometric methods.

Our paper also contributes to our understanding of what factors motivate firms to conduct splits. While traditional econometric models such as probit or logit regressions assume a linear relation between various features (such as size, book to market ratio, and current price level) and the probability of stock splits, ML methods do not impose a functional form between the target and features. Therefore, the feature importance results in this paper can potentially shed light on what features are important in predicting stock splits.

We also contribute to the literature in terms of methodology. While oversampling is widely used in statistics, it is not caught on in the finance literature. We show that when predicting rare corporate events, oversampling helps improve performance. Further, we show how to choose the optimal proportion of the minority class to oversample to achieve the best performance, by using grid search and a costed error function.

The remainder of the paper is organized as follows. We discuss the related literature in Section 2 and sample selection and data in Section 3. Section 4 introduces the two ensemble ML models and hyperparameter tuning. Section 5 discusses the results of predicting next-quarter splits. Section 6 provides results of predicting splits in the next year. Section 7 performs robustness checks on our portfolio return results. Section 8 concludes.

2. Literature review

Our study is related to a vast literature on stock splits. Many studies have documented positive stock returns before, immediately around, and after stock-split

announcements. Examples include, but are not limited to, Fama et al. (1969), Grinblatt, Masulis, and Titman (1984), Lakonishok and Lev (1987), Lamoureux and Poon (1987), Asquith, Healey, and Palepu (1989), McNichols and Dravid (1990), Maloney and Mulherin (1992), and Ikenberry, Rankine, and Stice (1996).

The most common reason for stock split cited in the literature is that stock splits increase the affordability and marketability of a firm's stock (Lamoureux and Poon, 1987; Maloney and Mulherin, 1992; Schultz, 2000).⁸ Brennan and Copeland (1988) argue that splits increase the bid-ask spread percentage, which in turn gives brokerage firms incentives to promote the stock and produce information on the firm. As a result, undervalued firms use splits to signal their value. Consistent with this argument, Chemmanur and Hu (2015) find that stock splits do lead to more information production by institutional investors.

Baker, Greenwood, and Wurgler (2009) argue that firms cater to the market demand for low nominal prices by conducting more splits in periods when low-priced stocks have a greater valuation premium than high-priced stocks. A recent study by Li and Ye (2021) models theoretically and shows empirically that firms set their nominal stock price at an optimal level to balance the minimum tick size constraint and the minimum lot size constraint to increase liquidity. The features we use in this paper to predict stocks are based on these studies.

Traditional econometric modeling uses probit or logit regressions to make predictions in finance (e.g., Billet and Xue, 2007, and Cremers, Nair, and John, 2009, on predicting mergers and acquisitions; Dimmock and Gerken, 2012, and Bollen and Pool,

⁸ By examining mutual fund share splits, Fernando, Krishnamurthy, and Spindt (1999) also find that share price levels are related to the marketability of mutual funds.

2012 on fraud predictions). With the rise of big data and the development of machine learning techniques in recent years (Goldstein, Spatt, and Ye, 2021), more studies have applied ML to finance research. For example, Erel et al. (2021) find that several machine learning methods perform better than humans in selecting board directors. Li et al. (2021) use a machine learning method called the word embedding model to extract measures of corporate culture from earnings call transcripts and find that the measures are correlated with business outcomes.

Machine learning methods have also been applied to other areas of economics and finance research in recent years: examples in asset pricing include Rossi (2018), Ke, Kelly, and Xiu (2019), and Giglio, Liao, and Xu (2021); examples in market microstructure include Easley et al. (2021) and Anand et al. (2021). Our paper adds to the literature by investigating how different ensemble machine learning methods perform in predicting stock splits.

3. Data, sample, and variables

3.1. Sample

Our sample uses all non-financial and non-utility stocks in CRSP. We use CUSIP identifiers that end with either 10 or 11 to make sure that only common stocks are included. Institutional holdings data are obtained from Thomson Financial's CDA/Spectrum 13F filings. Accounting information such as the book value of equity and the value of total assets is obtained from Compustat. Firm age and stock return information are from CRSP. To avoid market microstructure issues, we require the stock price at the end of the calendar quarter to be greater than \$5 per share.

Following the literature (e.g., Grinblatt, Masulis, and Titman, 1984; Li and Ye, 2021), we define a stock split as an observation with a split factor greater than or equal to 0.25.⁹ Stock dividends with a split factor less than 0.25 are not included in our stock-split sample because these events do not lead to a big enough change in the nominal share price to induce a large change in announcement returns, which may affect the power of the tests in the portfolio strategy part of the paper.

About 8% of the stock splits in CRSP have the execution date but not the announcement date. Since we are predicting stock split announcements, it's important to include these splits in our sample. If we omit them, they'll be incorrectly classified as non-splitters. We calculate the median number of days between the split announcement and execution in the non-missing sample as *median_gap*. We then fill in the missing announcement dates as the execution dates minus *median_gap*. The filled-in announcement dates may not exactly match the actual announcement dates. Since we are predicting whether a firm will announce a stock split in the next quarter or next year, not the exact announcement date, this is unlikely to cause any serious issues with our predictions.

3.2. Construction of variables

To measure the probability of stock splits in a calendar quarter, we use the split announcement date variable (*dclrdt*) in CRSP. If stock *i* announces a split in quarter *t*, the variable *split_q_{i,t}* has a value 1; otherwise, it has a value of 0. For example, if a firm

⁹ Split factor is defined as the number of additional shares per existing share the firm distributes to shareholders; for example, a two-for-one stock split (the most common stock splits) has a split factor of 1 and a three-for-two split has a split factor of 0.5. We include events with distribution codes of 5523, 5533, 5543, or 5552.

announces a stock split on January 10, 2000, the variable $split_q_{i,t}$ has a value of 1 in the calendar quarter 2000Q1, which spans from January 1 to March 31 of 2000.

Following the previous literature (e.g., Baker, Greenwood, and Wurgler, 2009; Perez, Shkilko, and Tang, 2011), we use twelve variables to predict stock splits. Specifically, to predict stock splits in the next quarter, the prediction model is as follows:

$$Prob(split_q_{i,t+1}=1) = f(Z_{i,t}), \quad (1)$$

where $f()$ stands for one of the three prediction models used in this paper: logit regressions, random forests, or gradient boost machines. $Z_{i,t}$ is a vector of twelve characteristics used to predict stock splits, and we will explain them one by one below. We also test the ability of the three predictive models to predict stock splits in the next *four* quarters (i.e., next year). In other words, we also check the following predictive models:

$$Prob(split_yr_{i,t+1}=1) = f(Z_{i,t}). \quad (3)$$

If stock i announces a split in any of the next four quarters, the variable $split_yr_{i,t+1}$ has a value 1; otherwise, it has a value of 0.

The twelve right-hand-side variables (i.e., features) we use to predict stock splits are defined as follows:

- $PRC_{i,t}$ is firm i 's price level at the end of quarter t , obtained from CRSP monthly stock file.
- $atq_{i,t}$ is firm i 's size at the end of quarter t , as measured by the value of total assets (in millions of dollars) from Compustat. To avoid look-ahead bias and make the investment portfolios implementable, we obtain total assets based on the most recent quarterly financial statement report date (Compustat variable rdq) before the

end of quarter t .¹⁰ The value of $atq_{i,t}$ is winsorized at the 1% and 99% levels at each quarter.

- $B/M_{i,t}$ is firm i 's book-to-market ratio of equity in quarter t . Book value of equity is based on Compustat variable $ceqq$, and market value of equity is the product of price per share and the number of shares outstanding (Compustat variables $prccq$ and $cshoq$), based on the most recent quarterly financial statement report date before the end of quarter t . We winsorize the value of $B/M_{i,t}$ at the 1% and 99% levels at each quarter.
- $pplast_{i,t}$ is the ratio of the current price ($PRC_{i,t}$ as defined above) to the price at last split. If a stock has never split before, we use the first price available for the stock in the CRSP daily stock file.
- $Num_splits_{i,t}$ is the number of times the stock has split in the past ten years. If a stock has never split before, the value is 0.
- $pct_split_{i,t}$ is the percentage of firms in stock i 's industry that have conducted at least one split in the past year. We use the 48 industries as defined in Fama and French (1997).
- $age_{i,t}$ is the age of stock i at the end of quarter t , as measured by the number of months the stock has appeared in CRSP.
- $mom_{i,t}$ is firm i 's twelve-month cumulative return ending at the last month of quarter t , winsorized at the 1% and 99% levels.

¹⁰ Throughout the paper, if quarterly financial statement report date (Compustat variable rdq) is missing, we assume it is 45 days after the fiscal quarter ending date (Compustat variable $datadate$). Since the SEC requires firms to file their 10-Q reports no later than 45 days after the end of the quarter, this avoids potential look-ahead bias in our results.

- $ios_{i,t}$ is firm i 's institutional ownership in quarter t . Since there is on average a 45-day lag in the reporting of 13F filings, we use one quarter lag to avoid look-ahead bias. For example, to obtain institutional ownership for quarter ending December 31, 2000, we use the filing date of September 30, 2000, in 13F filings. If a firm does not have institutional ownership information in 13F filings in a quarter, we set the value to 0.
- $LIQ_{i,t}$ is firm i 's liquidity in quarter t , measured as the absolute daily return divided by the daily volume of trade (in millions), as in Amihud (2002). We winsorize the value of $LIQ_{i,t}$ at the 1% and 99% levels each quarter.
- $VOL_{i,t}$ is the standard deviation of firm i ' daily returns in quarter t . Li and Ye (2021) find that volatility affects a firm's decision to split. We winsorize the value of $VOL_{i,t}$ at the 1% and 99% levels each quarter.
- P_t^{CME} is the log difference in the average market-to-book ratios of low- and high-priced stocks in quarter t , following Baker, Greenwood, and Wurgler (2009). Specifically, in quarter t , we divide all firms into terciles based on $PRC_{i,t}$, firm i 's price level at the end of quarter t . We then calculate the average market-to-book ratio of firms in each tercile. P_t^{CME} is the log difference in the average market-to-book ratios in low- and high-priced terciles. This variable captures the value premium for low priced stocks over high priced stocks in quarter t . Firms are more likely to split in periods when the value of P_t^{CME} is higher based on the catering theory proposed by Baker, Greenwood, and Wurgler (2009).

3.3. Summary statistics

We require the announcement date of the stock split to be in year 1980 or later because 13F starts to provide institutional ownership information in 1980. The final sample ranges from 1980Q1 to 2020Q4, with 375,164 observations. Panel A of Table 1 provides the number of stock splits used in the paper over time based on the year of announcement.

[Insert Table 1 about here]

There are a total of 7,210 stock splits in our sample. The year 1983 has the greatest number of observations by announcement date (556), whereas the year 2019 has the smallest number of stock-split announcements (5). We notice a trend of declining stock splits over time. While the average number of stock splits in the first five years of the sample is over 200, the average number of stock splits in the last five years of the sample is below 15.¹¹ To verify that there is indeed such a trend in the sample period, we also look at the mean and median stock prices over time. The results are reported in Panel B of Table 1.

In the first five years of the sample, the average nominal stock price is slightly above \$20 a share. In contrast, the average stock price in the last five years of the sample is close to \$200. One concern about average nominal price levels is that they are highly skewed due to a few outliers (such as Berkshire Hathaway and Google, whose stock prices at the end of 2020 were \$281,500/share and \$1,751.88/share, respectively). We, therefore, look at the median price levels each year as well. While the median price levels in the late 1980s are around \$15/share, the value increases to almost \$30/share in the late few years

¹¹ Da, Fang, and Lin (2022) also documents a decline in the number of stock splits in recent years. They show that the decline is related to the ability of retail investors to trade less than a full share starting in 2019.

of the sample. So there is indeed a trend over time in the sample period of increased nominal price levels.¹²

[Insert Table 2 about here]

Table 2 provides the summary statistics of the variables used in the paper. In the sample period, the average value of *split_q1* is 0.018181, indicating that the unconditional probability of a stock split in any given quarter is about 1.82%. The average probability of a stock split in the next four quarters is 6.6342%, based on the mean value of the variable of *split_yr*.

Note that many machine learning techniques such as deep neural networks require the features to be scaled (by either standardizing the feature to a standard normal distribution or using a min-max scaling so that the values are in the range of -1 to 1). Since both our ensemble ML methods are decision tree based, no scaling is needed because decision trees are not sensitive to the variance of the features. Indeed, scaling may actually increase the number of possible values in a feature, which complicates the decision tree node-splitting process.

4. Prediction models and hyperparameter tuning

We will train three different prediction models and then use them to make out-of-sample predictions on stock splits: logit regressions (Logit), random forests (RF), and gradient boosting machines (GBM). While logit regressions are widely used in finance

¹² As discussed in Section 3, we have included two features that can help predict the declining time trend in our prediction models: the percentage of firms in the stock's industry that have conducted at least one split in the past year and the log difference in the average market-to-book ratios of low- and high-priced stocks in the quarter.

research, the two ensemble ML methods are relatively new to finance. We therefore briefly introduce them below.

Further, in ML models, there are many hyperparameters one can choose.¹³ Sometimes, even if one chooses a good ML method, if the parameters are not optimized, the prediction models still do not work well. There is no hyperparameter tuning needed for logit regressions, except that we set the maximum number of iterations to 5,000 to make sure that the estimated parameter values converge. For RF and GBM, we will use grid search to find the best hyperparameter combinations that produce the best performance in predicting splits in the next quarter.¹⁴

4.1. Random Forests (RFs)

To understand random forests, we first need to understand decision trees. A decision tree is a Machine Learning classification method to predict target values (in our case, 1 for a stock split and 0 otherwise) based on input features. An internal node will have further sub-branches while a leaf has no further sub-branches. Each internal non-leaf node is labelled with one of the features to predict the outcome.¹⁵

The decision tree is known for being prone to overfitting: it works well on the training dataset, but poorly on the testing dataset. The idea behind random forest is to use

¹³ Hyperparameters are the parameters that control the learning process in ML models. Their values are set before the learning process starts and will not change due to training. In contrast, non-hyperparameters are derived from the training process.

¹⁴ In grid search, one uses trial and error to see which combination of hyperparameters lead to the best model performance. Since grid search is extremely computationally costly due to “curse of dimensionality,” it is impossible to cover all hyperparameter combinations. As a result, it is very likely that some hyperparameter combinations have better performance than the model used in this paper. This likelihood strengthens the conclusion of the paper that ML techniques may be better at predicting corporate events than traditional econometric methods.

¹⁵ For more information about decision trees, see, e.g., the Wikipedia post here https://en.wikipedia.org/wiki/Decision_tree_learning. The post uses the probability of survival of passengers on Titanic to show how the model uses decision tree to predict step by step based on the values of each of the following features: gender, age, number of spouses or siblings aboard.

multiple shallow trees instead of one deep tree to take advantage of the wisdom of the crowd to overcome the overfitting problem. In particular, RFs use an ensemble method called “bagging:” aggregating predictions from individual trees to form a consensus. RFs use the average predictions from the ensemble to make predictions.

The hyperparameters that we try to optimize are the following:

- *n_estimators*: the number of trees in the forest. This is the number of decision trees used to train the model, and we search over three different values: 50, 100, and 200.
- *min_samples_split*: the minimum number of observations required to split an internal node. We search over the values of 3, 5, and 10. For example, if the value of *min_samples_split* is set at 5, and there are 10 samples at an internal node, the split is allowed. In contrast, if there are 2 samples at another node, the split is not allowed.
- *min_samples_leaf*: the minimum number of observations required to be at a leaf node. We search over the three values of 3, 5, and 10. The model checks this hyperparameter before splitting the sample into leaves. For example, if the value of *min_samples_leaf* is set at 10, and the model is about to split 30 samples into two leaves with 8 and 22 samples each, the split is avoided (because the leaf with 8 samples has less than the cutoff value *min_samples_leaf* = 10).
- *max_depth*: the maximum number of levels of branches allowed in each decision tree. We search over the same three values: 3, 5, and 10. For example, if we set *max_depth* to 3, each decision tree can have at most three levels of branches.

We try to find the best combination of the above hyperparameters by using the training data to avoid the look-ahead bias and make the investment strategy in the paper

implementable. To avoid overfitting, we use k -fold cross-validation to find the best hyperparameters. Specifically, we split the training data into $k = 3$ subsets. We then iteratively fit the model 3 times: each time we train the model on $2/3$ of the data and evaluate the performance on the remaining $1/3$. We then average the performance of the model over the 3 iterations.

The optimized hyperparameters are: $n_estimators = 50$, $min_samples_split = 3$, $min_samples_leaf = 3$, and $max_depth = 3$. These are the hyperparameters that we'll use in RFs throughout the paper.

4.2. Gradient Boosting Machine (GBM)

GBM is another ensemble ML technique. Unlike the RF method we discussed above, it uses a method called “boosting” instead of “bagging” to improve performance. It uses an ensemble of weak prediction models (in our case, decision trees, which we have discussed above) to build a stronger model in a stage-wise manner. The idea is to see if a weaker learning mechanism can be modified to become better. GBMs perform better than other ML techniques in many cases. Even when they don't outperform other techniques, hyperparameter-optimized GBM models usually don't underperform by much. Some even call it the go-to ML technique if you are not sure which ML model to use.

In the case of predicting stock splits, the GBM method works as follows. In the first stage, it assigns a probability of stock split to all samples, using the sample mean (in the case of predicting next quarter splits, the value is 0.018181 based on the value in Table 2). It then calculates the residual for each sample: for the stock-split sample, the residual value is $1 - 0.018181 = 0.981819$; for the non-stock-split sample, the residual value is $0 - 0.018181 = -0.018181$. It then constructs a decision tree model to predict the residual value

for each observation. After the prediction, the model calculates the updated residual as the original residual plus the learning rate times the prediction from the decision tree.¹⁶ The model repeats the above process in the next stage. After the prespecified number of stages, the model is considered trained.

The hyperparameters that we try to optimize are the following:

- *n_stages*: the number of boosting stages to perform. More stages are usually better at reducing overfitting, but it takes longer to train the model. We search over two different values: 200 and 1000.
- *min_samples_split*: the minimum number of observations required to split an internal node in each decision tree, as we have explained above when discussing the RF model. We search over the values of 5 and 7.
- *min_samples_leaf*: the minimum number of observations required to be at a leaf node in a decision tree, as explained above in the RF model. We search over the values of 3 and 5.
- *max_depth*: the maximum number of levels of branches allowed in each decision tree. We search over the values of 10 and 12.

We try to find the best combination of the above hyperparameters by using the training data to avoid the look-ahead bias and make the investment strategy in the paper implementable. Similar to what we have done with the RF method, we use k-fold cross-validation to find the best hyperparameters, where $k=3$. The optimized hyperparameters

¹⁶ The learning rate is a hyperparameter that controls how fast the model updates the prediction in each stage. The value of the learning rate is between 0 and 1. If the value is set too low, it takes a long time for the model to converge. If the value is set too high, the model may be influenced too much by outliers and the trained model may overfit. We set the learning rate at 0.1 in this paper.

are: $n_stages = 1000$, $min_samples_split = 5$, $min_samples_leaf = 3$, and $max_depth = 12$.

These are the hyperparameters that we'll use in GBMs throughout the paper.

5. Predict stock splits in the next quarter

In this section, we compare the performance of the three models in predicting stock splits in the next quarter. There are 41 years of data from 1980Q1 to 2020Q4. We use the first 82 quarters (1980Q1 to 2000Q2) as the training data and the last 82 quarters (2000Q3 to 2020Q4) as the out-of-sample testing data.

Since stock splits are rare events, accuracy may not be the best measure of performance. For example, in a firm quarter, the unconditional probability of a stock split is 1.82% based on the average value of *split_q1* from Table 2. If one predicts no-split for all firm quarters, the accuracy would be 98.18%, an extremely high value. However, one wouldn't learn anything from the prediction despite the high accuracy.

Instead, we'll look at other measures such as the ROC curve and the AUC score, which are explained below.

5.1. The ROC curve and the AUC score

To evaluate the performance of a prediction model with imbalanced data, the literature usually uses the receiver operating characteristic (ROC) curve. The ROC curve plots the true positive rate against the false positive rate and shows the performance of the model at various false positive rate cutoff points. Let's use TP to denote the number of true positives, TN the number of true negatives, FP the number of false positives, and FN the number of false negatives. Then true positive rate and false positive rate are defined as follows:

$$\text{True Positive Rate} = \frac{TP}{TP + FN}$$

$$\text{False Positive Rate} = \frac{FP}{FP + TN}$$

The area under the receiver operating characteristic curve (AUC) score can be interpreted as the probability that a prediction model ranks a true positive outcome higher than a true negative outcome.

[Insert Figure 1 about here]

Figure 1 shows the ROC curves and the AUC scores of the three prediction models used in the paper when predicting stock splits in the next quarter. Both GBM and RF have an AUC score of about 0.85. In contrast, Logit has an AUC score of around 0.75. The results show that the two ensemble ML models perform better than the traditional econometric method of logistic regressions.

5.2. Predicting splits by allowing 5% ex ante false positive rate

The three prediction models generate a probability of split for each observation. We need to use a cutoff value to classify an observation as a splitter or a nonsplitter. A high AUC score indicates that for relatively low false positive rates, the model can correctly identify firms that will indeed split their stocks in the next quarter. However, our purpose here is not necessarily predicting as many true stock splits as possible. Instead, we want to form a portfolio with high future abnormal returns. For that purpose, we need the number of false positive predictions to be low as well.

In our setting, there is a tradeoff between correctly predicting stock splits and falsely classifying a nonsplitter as a splitter. The false positive rate can be interpreted as the opportunity cost of investors incorrectly investing their money in stocks without

abnormal returns, which can be costly in our case. Failing to predict a true splitter is costly as well: it may reduce the expected abnormal return and increase the return volatility on the portfolio if the investment strategy is to buy all positive predictions.

Similar to many papers in the literature (e.g, Dimmock and Gerken, 2012), we allow for a 5% ex ante false positive rate when classifying a binary event. To calculate the ex-ante false positive rate, we first look at the predicted values among all nonsplitters in the training dataset (clean sample) and use the 95 percentile values as the cutoff values. We then look at the predicted values in the testing dataset and classify observations with predicted values above the cutoff as positives (i.e., splitters), and others as negatives (i.e., nonsplitters).

[Insert Table 3 about here]

Panel A of Table 3 reports the number of true positives, false positives, and so on for the three prediction models if we allow for a 5% ex-ante false positive rate. Results show that GBM has the highest number of true positive predictions, at 968. RF has 562 true positives, while Logit has just 147. However, GBM also has the highest number of false positive predictions, at 34,752. RF has 11,724, while Logit has 6,631 false positive predictions.

Since a portfolio with most true positives and least false negatives is likely to have the highest abnormal return, results in Table 3 provide no indication of which model will generate the highest abnormal returns (if any) if we buy all stocks that a model predicts to split. We'll investigate that empirically soon.

5.3. Feature importance

One criticism of ML is that it acts like a black box and it is hard for us to understand which features help make the predictions. We have used twelve different variables to predict stock splits in the paper. To investigate what variables (i.e., features) are the most important ones in predicting splits, we perform feature importance testing in this section for the two ensemble ML techniques. Specifically, we examine Shapley Additive Explanations (SHAP) values of different features to understand their importance. The SHAP feature importance method is first developed by Lundberg and Lee (2017) and is used in some studies in finance (e.g., Erel et al., 2021). In a nutshell, SHAP values quantify the contribution of each feature in predicting the outcome, as well as whether the feature is positively or negatively related to the outcome. Specifically, it considers all possible combinations of available features used to predict the outcome. The marginal contribution of a feature is the improvement in the average prediction performance of the model with and without the feature.

[Insert Figure 2 about here]

Figure 2 is a bar-chart of feature importance of the twelve variables when the GBM model is used. The length of the bar shows the importance of the feature, with colors red and blue indicating a positive and negative impact on the predicted probability of split, respectively. Figure 2 shows that the price level is the most important factor in predicting splits, followed by the ratio of the price level to the price at last split. Both features increase a stock's probability to split. The third most important feature is the size of the stock. The blue color indicates that firm size reduces a stock's propensity to split, which is consistent with anecdotal evidence that larger firms are more likely to keep their prices at a higher

level than smaller firms do (e.g., Berkshire Hathaway and Google). The fourth important feature is the return of the stock in the past 12 months, which increases the probability that a stock splits. After the first four features, there is a sharp drop off in feature importance for the other eight features.

[Insert Figure 3 about here]

A bar-chart of the feature importance of the twelve variables when RF is used is shown in Figure 3. Similar to Figure 2, results show that the ratio of the current price level to the price at last split, the price level, and return of the stock in the past 12 months are three important features that increase the probability of a split. But unlike in Figure 2, firm size is not a very important factor in predicting splits. Book to market ratio plays a somewhat important role in negatively affecting the probability of splits. Other features do not seem to be important in predicting splits.

In summary, both GBM and RF show that the stock price level, the ratio of the current price to the price at last split, and the return on the stock in the past 12 months, are important in predicting stock splits.

5.4. Portfolio returns

Based on each prediction model, we form an investment strategy in which we purchase all positive predictions. We then assess the abnormal portfolio returns based on CAPM, the Fama-French three-factor model, or the Fama-French five-factor model. In rare cases where there are no positive predictions in a quarter, we assume the money is invested in the market index (that is, the return on the factor MKT from Ken French's monthly factor returns data).¹⁷

¹⁷ The factor returns are obtained from Ken French's website https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

We form both equal-weighted and value-weighted portfolios. Since predictions are made at the end of each quarter, value-weighted portfolios are rebalanced every quarter (that is, form a value-weighted portfolio at quarter beginning and hold for 3 months; sell all of them after 3 months and form a new portfolio based on the new quarterly predictions). In contrast, equal-weighted portfolios are rebalanced every month. Specifically, we form an equal-weighted portfolio at quarter beginning and rebalance each month to keep the portfolio equal-weighted; sell all of them after 3 months and form a new portfolio based on the new quarterly predictions.

Table 4 reports the portfolio alphas based on the Fama-French five-factor model. Results based on CAPM or the Fama-French three-factor model are similar and we will show some of them later when performing robustness tests.

[Insert Table 4 about here]

In Panel A, the predictions are made for the three models by allowing for a 5% ex-ante false positive rate. We then regress the monthly returns of the equal-weighted portfolios of all positive predictions on the five factors and report the results. The t-statistics based on Newey-West standard errors with one lag are reported in parentheses under the coefficients. We use the Newey-West standard errors with one lag to account for the possibility that the returns on a stock are correlated over time.¹⁸ The abnormal portfolio return, alpha, is the intercept of the regressions. Results show no significant alpha for any of the three prediction models. The alphas are actually negative, albeit insignificant. In Panel B, we repeat what we have done in Panel A but form value-weighted instead of

¹⁸ Results are close to identical when we use two or three lags.

equal-weighted portfolios. None of the portfolios formed by the three prediction models can generate abnormal returns, either.

In summary, the results in Table 4 show that none of the three prediction models can generate abnormal returns, despite the high AUC scores from the two ensemble ML techniques. We will examine the reasons below.

5.5. Why do high AUC scores not lead to high portfolio returns?

The two ensemble ML models generate very high AUC scores, both at 0.85. Why do the positive predictions from the two models not generate abnormal returns?

One possibility is that the abnormal returns around stock splits are not high enough in the testing period. To investigate if this is indeed the case, we look at the average monthly returns of all stock splits 11 months before and 11 months after the split announcement in the testing period of July 1, 2000, to December 31, 2020. Results are reported in Table 5. We also plot the average monthly returns over time in Figure 4.

[Insert Table 5 and Figure 4 about here]

Table 5 shows that in the eleven months before split announcements, (i.e., from month -11 to month -1), the average monthly stock returns of the splitters range from 4.21% to 6.01%. The average return in the announcement month is 6.42%. The returns in the eleven months after split announcements are much lower: the average monthly return ranges from 0.66% to 1.92%. The pattern is also shown in Figure 4.

Therefore, the lack of abnormal returns from the prediction models is unlikely due to low returns around stock splits. The more likely reason for low portfolio returns is the fact that stock splits are very rare events, which leads to an extremely imbalanced sample

in which there are many more non-splitters than splitters. As a result, there are too many false positives and not enough true positives in our portfolio to generate significant alphas.

We will use two methods to overcome this issue. First, we will use SMOTE oversampling to increase the instances of splitters in the sample when training the model. Second, we will use the two ensemble ML techniques to predict stock splits in the next *four* quarters (that is, next year) instead of in the next quarter and see if they can generate abnormal portfolio returns. The unconditional probability of a split in the next four quarters is 6.6342% in our sample, much higher than the unconditional probability of a split in the next quarter, which is only 1.8181%. As a result, the ratio of true positives to false positives is greater compared to the portfolios formed based on predictions for stock splits in the next quarter. The prediction will be more precise. Additionally, predicting stock splits in the next year captures more months of return before split announcements. E.g., in the case that a portfolio is formed on January 1 and a split is announced on October 1, the portfolio captures the stock return of the splitter from January to September. As shown in Table 5, splitters have higher returns in the months before announcements. Therefore, predicting splits in the next year should increase the likelihood that the prediction portfolios generate alphas.

5.6. Use SMOTE oversampling to overcome the rare event problem

Oversampling is an effective tool to improve the performance of prediction models when the sample is extremely imbalanced (e.g., the number of observations of one class is smaller than 5% of that of the other class). In our sample, splitters represent 1.82% of all

observations, so oversampling should improve performance in our setting. We use the SMOTE oversampling method developed by Chawla et al. (2002).¹⁹

To determine what proportion of splitters should appear in the oversampled training dataset, we follow the literature (e.g., Khan et al., 2015) and use a costed error function to determine which proportion works best. The costed error function assigns different penalties to false positives and false negatives based on how costly they are. In our setting, false negatives are costly, but not as costly as false positives are. A false positive means that investors incorrectly invest their money in stocks without abnormal returns, which can be extremely costly in our case because it dilutes portfolio returns. Failing to predict a true splitter is not as costly in our case because, as long as the total number of true positive predictions is at least one, the portfolio will have abnormal returns that are generated by split announcements.

For this reason, we define the costed error function as follows:

$$\text{costed error function} = \frac{1}{N} \sum_{i=1}^N \text{cost}_i$$

where

$$\text{cost}_i = \begin{cases} 20, & \text{if } i \text{ is a false positive} \\ 1, & \text{if } i \text{ is a false negative} \\ 0, & \text{otherwise} \end{cases}$$

That is, we assign a cost of 20 to a false positive prediction, 1 to a false negative prediction, and 0 to true positives and true negatives.²⁰

¹⁹ The *imblearn* library in Python provides a convenient way to implement SMOTE oversampling. The details of the method can be found at the library's documentation page https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.SMOTE.html.

²⁰ We have similar results if we set the cost on false positives to 25 or 30.

We try to find the best proportion of oversampling by using the training data to avoid the look-ahead bias and make the investment strategy in the paper implementable. To avoid overfitting, we use k-fold cross-validation to find the best oversampling proportion. We split the training data into $k=3$ subsets and iteratively fit the model 3 times: each time we train the model on $2/3$ of the data and evaluate the performance on the remaining $1/3$. We then average the costed errors of the model over the 3 iterations. Specifically, we grid search five cases so that the ratio of splitters over nonsplitters in the training dataset is 0.1, 0.3, 0.5, 0.7, or 0.9. We then use the oversampled training dataset to fit each of the three prediction models and see which proportion produces the smallest costed error. We again allow a 5% ex ante false positive rate when classifying splitters and nonsplitters based on the predicted probabilities. When Logit, GBM, and RF are used, the lowest average costed errors are achieved at oversampling proportions of 0.9, 0.3, and 0.7, respectively. We will use these proportions to oversample the training dataset when training the three models. The trained models are then used to predict splits in the testing period.

[Insert Figure 5 about here]

Figure 5 shows the ROC curves and AUC scores in the testing dataset for the three models with oversampled training. All three models see increases in the AUC scores: for logit regressions, the AUC score has increased from 0.7547 to 0.7898; from 0.8478 to 0.8560 for GBM; from 0.8481 to 0.8655 for RF.

[Insert Table 6 about here]

The confusion matrixes for the three models are shown in Panel A of Table 6. Compared to the confusion matrixes in Table 3, we can see that for GBM, the number of

TP has increased from 968 to 1,290, while the number of FP increased from 34,752 to 38,182. The precision, which is defined as $TP/(TP+FP)$, has increased (from $968/(968+34,752)=2.71\%$ to $1,290/(1,290+38,182)=3.27\%$). For RF, the number of TP has decreased from 562 to 269, while the number of FP decreased from 11,724 to 2,953, resulting in a substantial increase in precision of the prediction (from $562/(562+11,724)=4.57\%$ to $269/(269+2,953)=8.35\%$). Next, we will see if the improvement in precision leads to abnormal portfolio returns.

Panel B shows the performance of the equal-weighted and value-weighted portfolios of positive predictions from the three models. The Fama and French (2015) five-factor model is used but other factor models produce similar results. None of the portfolios generate abnormal returns. Since small stocks usually have higher abnormal announcement returns due to greater degrees of information asymmetry, Panel C instead focuses on positive predictions of stocks in the smallest size quintile in the quarter of portfolio formation (i.e., when the predictions are made). Results show that the GBM equal-weighted portfolio has a monthly alpha of 0.2586%, statistically significant at the 5% level. However, the GBM value-weighted portfolio does not have abnormal returns. The RF equal-weighted portfolio has a monthly alpha of 0.9501%, statistically significant at the 5% level, while the RF value-weighted portfolio has a monthly alpha of 0.8099%, statistically significant at the 10% level.

To recap, SMOTE oversampling helps the performance of all three prediction models, and it leads to abnormal returns for GBM and RF predictions among small stocks.

6. Predict stock splits in the next year

In this section, we use the three models to predict stock splits in the next year and see if any of them can generate portfolios with abnormal returns. Similar to the previous section, we use the first 82 quarters (1980Q1 to 2000Q2) to train the model. We then use the trained model to predict stock splits from 2000Q3 to 2020Q4.²¹

6.1. AUC scores and ROC curves

Figure 6 shows the ROC curves of the three prediction models when the target variable is *split_yr*. GBM has an AUC score of 0.8624 while RF has an AUC score of 0.8617. The logit regression has an AUC score of 0.7800.

[Insert Figure 6 about here]

The AUC scores show that there is some improvement in the ability of the three models in predicting stock splits in the next year than in predicting splits in the next quarter. We will investigate whether we can translate the better predictions into abnormal portfolio returns.

6.2. Classify positive and negative predictions

Table 7 reports the number of true positives, false positives, and so on for the three prediction models if we allow for a 5% ex-ante false positive rate. That is, we look at the predicted values among all nonsplitters in the training dataset (clean sample) and use the 95 percentile values as the cutoffs. We then look at the predicted values in the testing dataset and classify observations with predicted values above the cutoffs as splitters, and others as nonsplitters.

²¹ Unreported results show that SMOTE oversampling does not help the performance of any of the three models when predicting splits in the next year. The unconditional probability of a split is 6.6342% in our sample, and this makes splits in the next year not an extremely rare event. This seems to indicate that oversampling only helps extremely imbalanced data.

[Insert Table 7 about here]

Results show that the three prediction models all have more positive predictions than those reported in Table 3 when we predict next-quarter stock splits. The patterns are also similar: GBM has the highest number of true positive predictions, at 4,411, but it also has the highest number of false positive predictions, at 79,570. The logit regression has the lowest number of both true positives and false positives among the three models. The RF method falls in between.

Next, we investigate if the portfolios formed based on positive predictions can generate abnormal returns.

6.3. Portfolio returns

For each prediction model, we implement an investment strategy of buying all positive predictions and holding them for twelve months. We then look at the abnormal return based on the Fama-French five-factor model.

Table 8 reports the Fama-French five-factor regression results for the three prediction models, based on both equal- and value-weighted portfolios.

[Insert Table 8 about here]

In Panel A, the predictions are based on the three models by allowing for a 5% ex-ante false positive rate. Equal-weighted portfolios are formed. Results show that the GBM portfolio has an abnormal monthly return of 0.1396%, statistically significant at the 5% level. Neither RF nor Logit regressions generate abnormal returns. In Panel B, we change the portfolio to value-weighted. The predictions are again based on the three models by allowing for a 5% ex-ante false positive rate. Results show that none of the three prediction models can generate abnormal returns.

In summary, the results in Table 8 show that GBM can generate abnormal returns if we form an equal-weighted portfolio. The fact that the value-weighted portfolio does not have abnormal returns based on GBM predictions suggests that the abnormal returns come mainly from small stocks. We will later use this observation to implement a strategy that generates larger alphas.

6.4. Why GBM predictions generate alphas while the other two do not?

The results in Table 8 indicate that the equal-weighted portfolio from GBM predictions generates abnormal returns. Next, we examine what is the source of this alpha. In particular, we look at the returns from true positives versus false positives separately and see if the alpha from the GBM prediction portfolio is indeed due to correctly predicting next-year stock splits.

Table 9 reports the Fama-French five-factor regression results for the three prediction models, based on both equal- and value-weighted portfolios of true positive and false positive predictions.

[Insert Table 9 about here]

In Panel A, we look at the true positive predictions from the three models and form equal-weighted portfolios. Note that this strategy is not implementable since we do not know which prediction will turn out to be a true positive or which one will be a false positive. We form the portfolio and look at the returns for ex post analysis purposes only. Results show that the true positive predictions from all three models generate large and significant monthly alphas: 1.0401% for Logit, 1.6705% for GBM, and 1.2663% for RF. The GBM model generates the highest alpha among true positive predictions.

In Panel B, we form value-weighted portfolios of true positive predictions from the three models. The GBM model again generates the highest alpha among the three models, with a 0.9691% monthly alpha. The RF model generates a 0.8891% monthly alpha. Surprisingly, the value-weighted portfolio of true positive predictions from the logit regression does not have a statistically significant alpha. Given the results in Table 5 on the high returns around stock splits, this result warrants more investigation. Therefore, we look at the alpha from CAMP and Fama French three-factor models of the portfolio in Panel C.

Results in Panel C show that the value-weighted portfolios of true positive predictions from Logit regressions do have statistically significant monthly alphas based on CAMP and Fama French three-factor models: at 0.9798% and 0.9666%, respectively. The reason that there is no significant alpha from the Fama French five-factor model is due to the large positive factor loadings on RMW and CMA, the profitability and investment factors. That is, the value-weighted portfolio of true positive predictions from Logit regressions does have high raw returns, but those returns are compensations for risks associated with the profitability and investment factors.

In Panels D and E, we form equal-weighted and value-weighted portfolios, respectively, of false positive predictions from the three models. None of the portfolios has a significant positive monthly alpha. In fact, the equal-weighted portfolios of false positive predictions from logit regressions even have significantly negative monthly alpha, at -0.2578%.

In summary, the results in Table 9 show that the abnormal portfolio returns from GBM predictions come indeed from correctly predicting stock splits.

6.5. Portfolios that generate higher alphas based on firm size

We have seen in Table 8 that the equal-weighted portfolio of GBM predictions has positive monthly alphas, while the value-weighted portfolio does not. This suggests that the abnormal returns on smaller stocks are greater than those on larger stocks. Based on this, we form portfolios of small stocks that can potentially generate higher alphas.

Specifically, stocks are sorted into quintiles based on their market capitalizations at the end of each quarter. Portfolios are formed to purchase all positive predictions in the smallest quintile at the end of the quarter and hold them for 12 months. Table 10 reports the Fama-French five-factor regression results for both equal- and value-weighted portfolios.

[Insert Table 10 about here]

In Panel A, equal-weighted portfolios are formed. The results show that GBM predictions among small stocks generate a monthly alpha of 0.3765%. The other two models do not generate abnormal returns even among small stocks. In Panel B, value-weighted portfolios are formed. The results show that GBM predictions among small stocks generate a monthly alpha of 0.3305%. The other two models do not generate abnormal returns. Abnormal returns are larger among small stocks could be due to the fact that small stocks tend to be less informationally efficient. Therefore, the likelihood that the firm will conduct a split in the near future is not incorporated in the price. When the firm announces a stock split, the market reaction tends to be stronger than that from large stocks.

In summary, Table 10 shows that if we invest only in small stocks, we can increase the abnormal returns from GBM predictions. Further, the strategy works with both equal- and value-weighted portfolios.

7. Robustness tests

Results in Section 6 show that GBM predictions for stock splits in the next year can generate abnormal portfolio returns based on the Fama-French five-factor model. In this section, we test whether the portfolio results are robust. We first test if our results hold if we exclude the year 2020 due to COVID-19 concerns. We also test if the portfolios have abnormal returns if we use CAPM, the Fama-French three-factor model, or if we add the momentum factor to the five-factor model.

Further, throughout the paper, we allow for a 5% ex ante false positive rate when classifying observations into positive and negative predictions. We will check if the results hold if we allow for a 1% ex ante false positive rate instead.

When we exclude year 2020 to take into the consideration of any effect of COVID-19 on the portfolio returns, the pattern is the same as those in Table 8. To save space, results are not included in the paper but they are available upon request from the authors.

So far we have only shown the portfolio alphas using the Fama-French five-factor model. In this section, we will show results using CAPM, the Fama-French three-factor model, or adding the momentum factor to the Fama-French five-factor model.

[Insert Table 11 about here]

In Panel A of Table 11, we report regression results for the equal-weighted portfolios based on GBM predictions. The first column shows results from CAPM, with a monthly abnormal return of 0.1870%, statistically significant at the 10% level. If we use the Fama-French three-factor model (as shown in the second column), the monthly abnormal return is not statistically significant. The last column shows results if we add the

momentum factor to the Fama-French five-factor model: the alpha increases from 0.1396% to 0.1397% and remains statistically significant at the 5% level. Note that the GBM prediction portfolio does not have a significant factor loading on the momentum factor. This indicates that the results are not due to the price run-ups before splits. In Panel B of Table 12, the alpha remains statistically insignificant for the GBM prediction portfolios in all four models.

We have seen in Table 10 that portfolios of small stocks among positive predictions from the GBM model have monthly alphas of 0.3765% (equal-weighted) and 0.3305% (value-weighted) based on the Fama-French five-factor model. Below we will test if the results hold qualitatively with other factor models.

Panels C and D in Table 11 show that GBM predictions among small stocks yield large and statistically significant monthly alphas based on other factor models. For example, Panel C shows that the equal-weighted portfolio has a monthly alpha of 0.3765% if we add the momentum factor to the Fama-French five-factor model. The alpha is statistically significant at the 1% level. Panel D shows that the value-weighted portfolio has a monthly alpha of 0.3980%, statistically significant at the 1% level, based on the Fama-French three-factor model. To recap, the results in Table 11 show that results on portfolio returns hold if we use other factor models instead of the Fama-French five-factor model.

When classifying splitters and nonsplitters, we allow for a 5% ex ante false positive rate. We examine if results change if a 1% ex ante false positive rate is used instead. Specifically, we first look at the predicted values among all nonsplitters in the training dataset (clean sample) and use the 99 percentile values as cutoffs. In the testing dataset, observations with predicted values above the cutoff are classified as splitters and others

nonsplitters. Unreported tables show that if we allow for a 1% ex ante false positive rate instead of 5% when classifying splitters and nonsplitters, the results remain similar. These tables are available upon request.

8. Conclusion

In this paper, we predict stock splits using two ensemble machine learning techniques, with the goal of forming implementable portfolios to generate abnormal returns. We find that when predicting stock splits in the next quarter, GBM and RF achieve area under the receiver operating characteristic curve (AUC) scores of around 0.8560 and 0.8655, respectively, with the help of SMOTE oversampling. GBM and RF predictions generate monthly five-factor alphas (Fama and French, 2015) of 0.26% and 0.95% among stocks in the smallest size quintile. Three important features for predicting stock splits in both ensemble ML methods are current price levels, the ratio of current price to the price at last split, and stock returns in the past twelve months. When predicting stock splits in the next year, oversampling does not improve performance, and GBMs generate monthly five-factor alphas of 0.38% among small stocks. Our results show that machine learning methods are promising in helping predict corporate events. When predicting rare corporate events, oversampling helps improve performance.

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Figure 1: ROC Curve for prediction of stock splits in the next quarter without oversampling

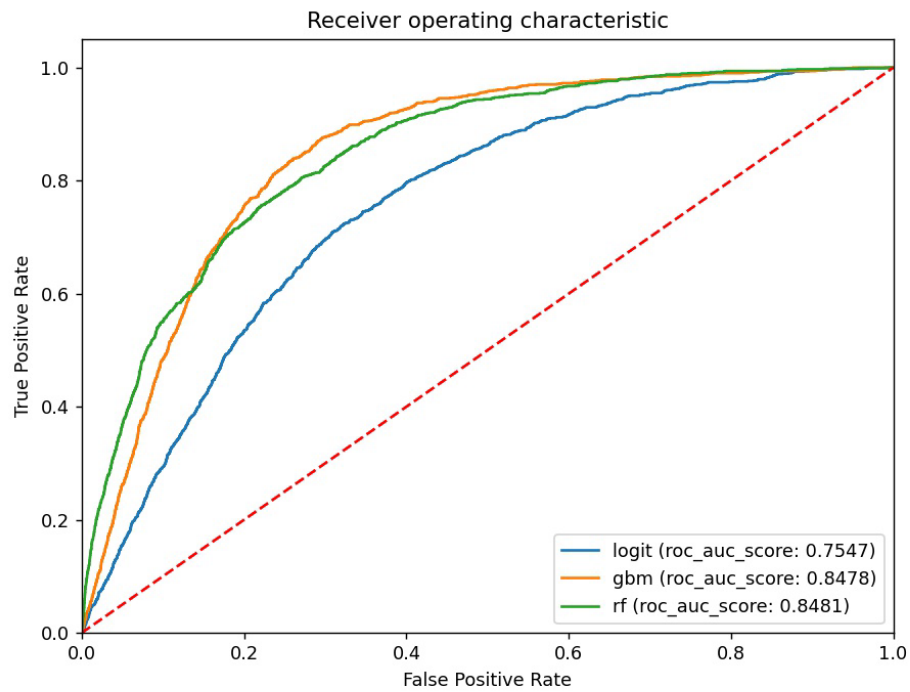


Figure 2: SHAP feature importance for the GBM model

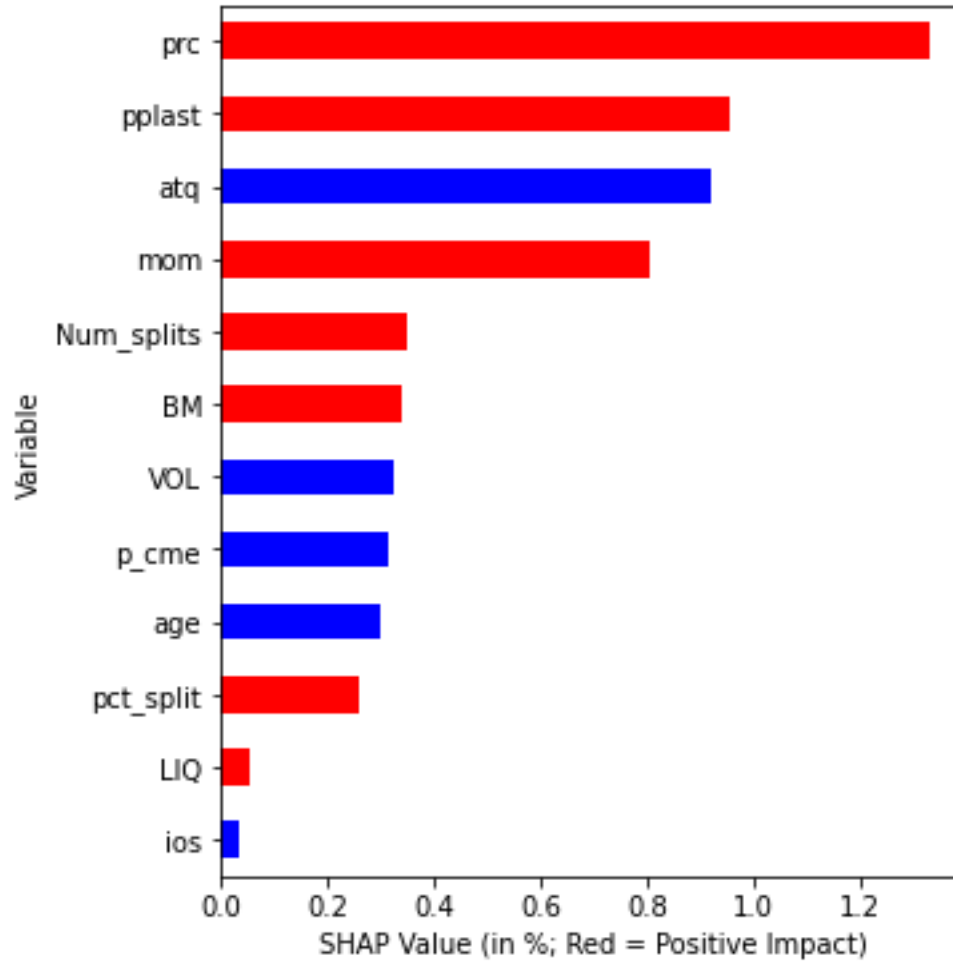


Figure 3: SHAP feature importance for the RF model

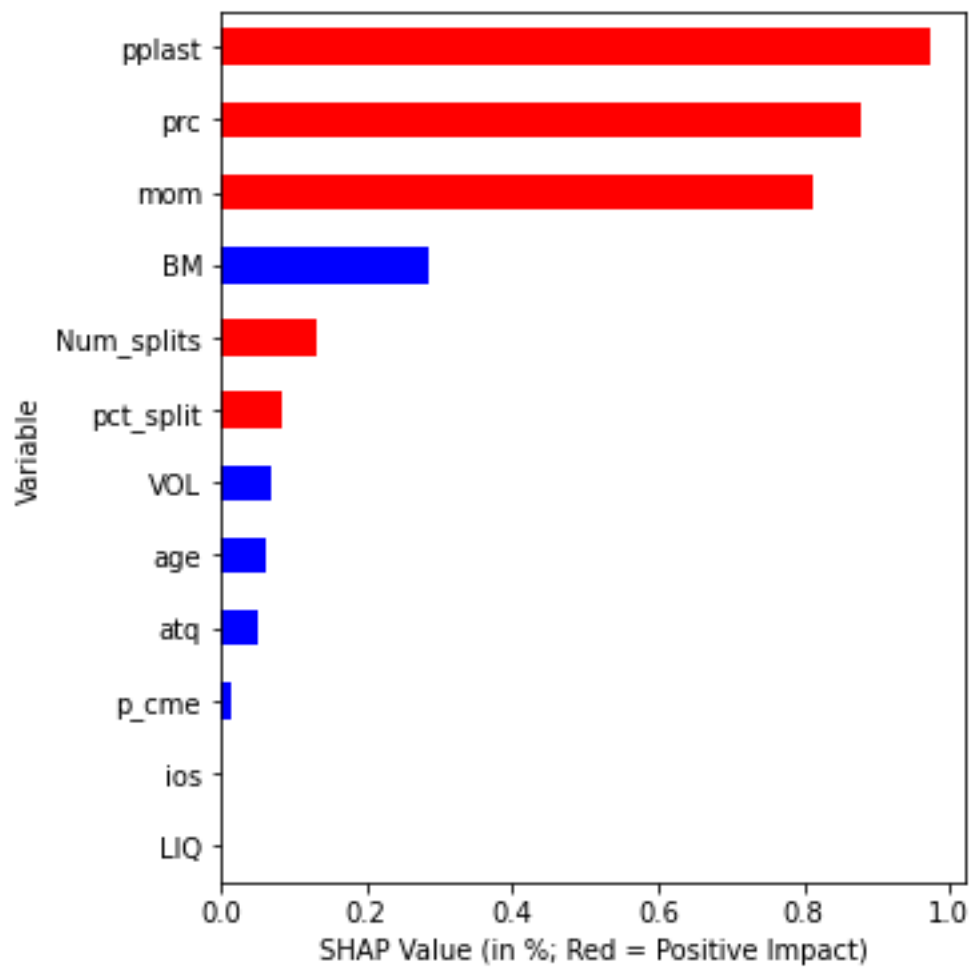


Figure 4: Average monthly returns around stock splits

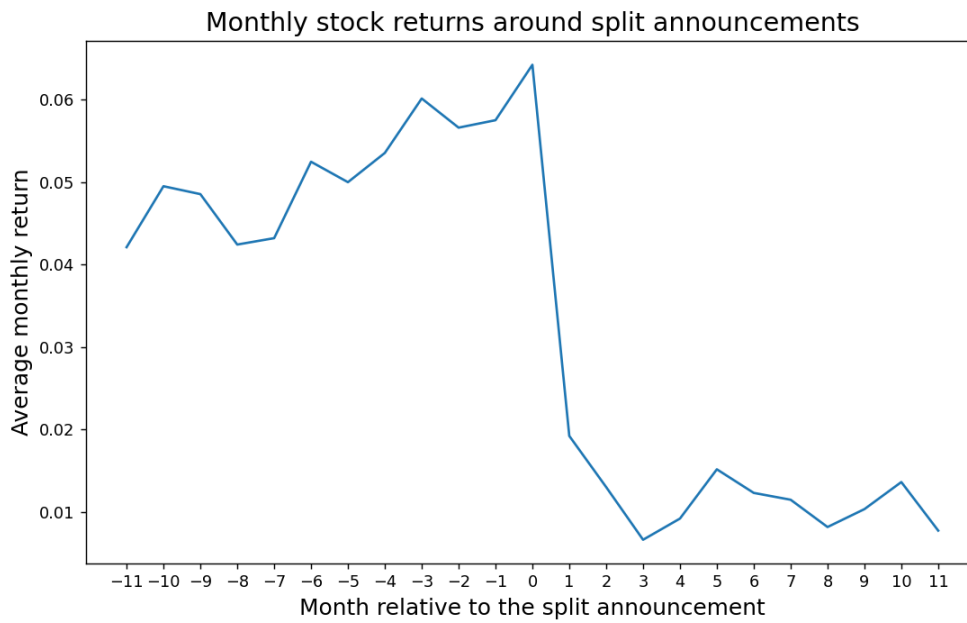


Figure 5: ROC Curve for prediction of stock splits in the next quarter with SMOTE oversampling

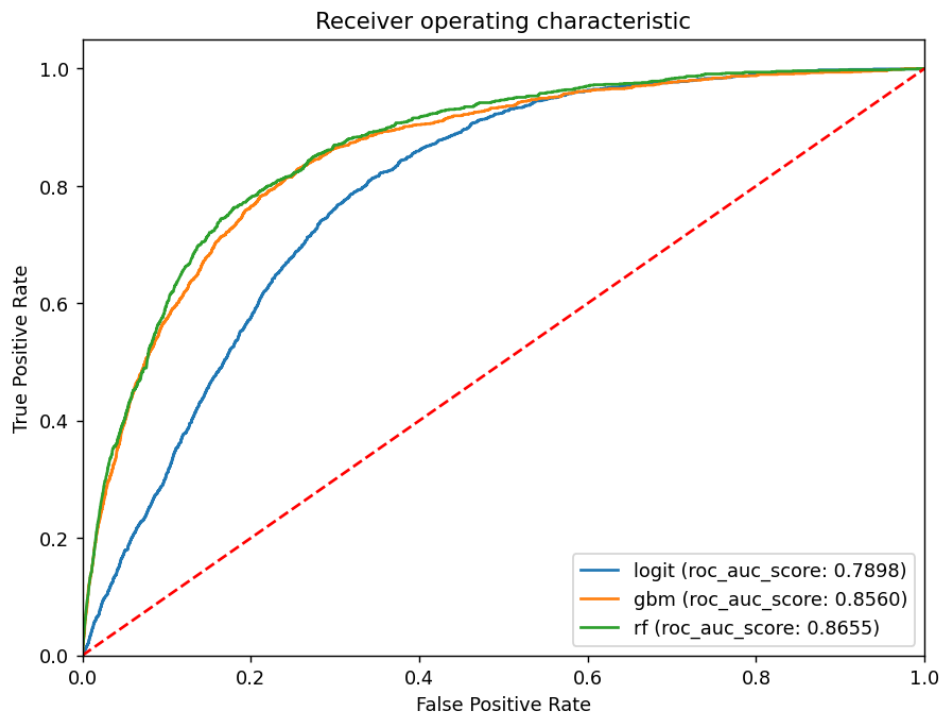


Figure 6: ROC Curve for prediction of stock splits in the next year without SMOTE oversampling

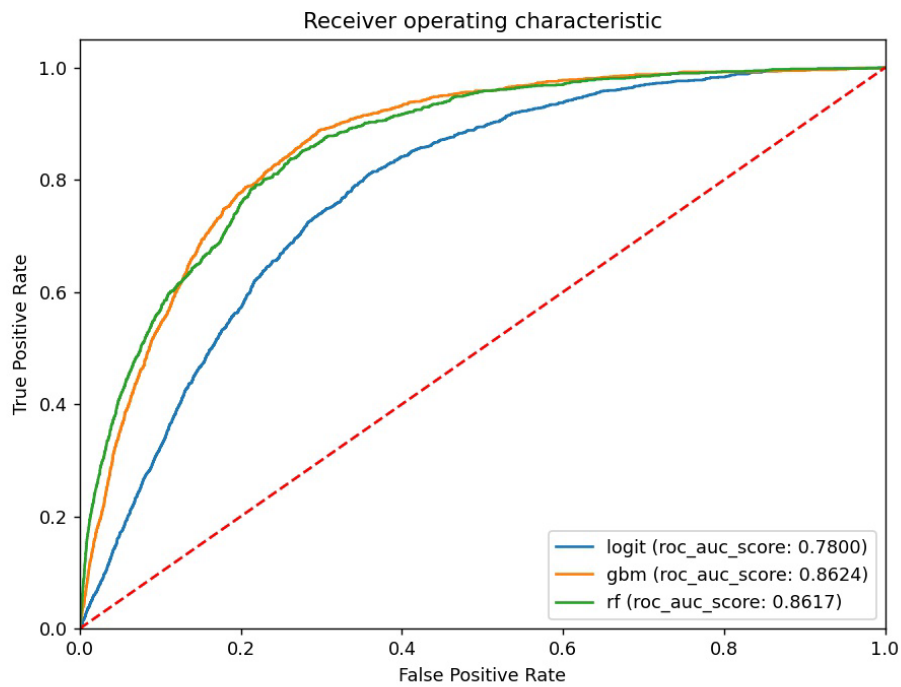


Table 1: The frequency of stock splits and nominal price levels over time

This table provides the distribution of stock splits on common stocks with split factors greater than or equal to 0.25 between 1980 and 2020. Financial and utility stocks are excluded. Panel A provides the number of firm-quarter observations and the number of split announcements in the year. Panel A provides the mean and median of price levels of all firms in the sample in the year.

Panel A: Number of stock split over time

Year	Number of firm quarters	Number of split announcements	Year	Number of firm quarters	Number of split announcements
1980	5,532	251	2001	9,697	108
1981	5,227	205	2002	8,984	106
1982	6,210	188	2003	9,391	132
1983	9,636	556	2004	10,109	171
1984	9,327	203	2005	9,777	186
1985	9,379	311	2006	9,896	137
1986	9,295	393	2007	9,735	109
1987	9,287	334	2008	8,391	35
1988	8,823	151	2009	7,857	8
1989	8,530	217	2010	8,298	42
1990	7,500	140	2011	8,086	43
1991	8,037	206	2012	8,024	40
1992	8,617	263	2013	8,298	41
1993	9,911	284	2014	8,515	38
1994	10,611	213	2015	8,598	30
1995	11,828	326	2016	8,551	13
1996	12,425	364	2017	8,537	18
1997	13,284	394	2018	8,434	13
1998	12,831	332	2019	8,348	5
1999	11,985	282	2020	8,232	13
2000	11,131	309			
Total				375,164	7,210

Panel B: Average price levels over time

Year	Mean	Median	Year	Mean	Median
1980	\$23.95	\$19.50	2001	\$51.31	\$17.31
1981	\$23.45	\$19.50	2002	\$53.68	\$16.80
1982	\$20.72	\$16.88	2003	\$54.11	\$17.38
1983	\$23.71	\$18.50	2004	\$61.53	\$20.20
1984	\$20.26	\$15.38	2005	\$62.79	\$21.76
1985	\$22.07	\$15.88	2006	\$67.95	\$22.65
1986	\$23.90	\$16.25	2007	\$80.80	\$23.84
1987	\$23.84	\$15.88	2008	\$84.47	\$18.91
1988	\$22.14	\$14.63	2009	\$72.45	\$17.66
1989	\$25.10	\$15.50	2010	\$87.46	\$20.97
1990	\$24.17	\$14.75	2011	\$89.48	\$22.81
1991	\$26.28	\$16.00	2012	\$97.86	\$23.54
1992	\$25.73	\$15.75	2013	\$121.28	\$27.70
1993	\$27.56	\$15.88	2014	\$140.23	\$30.24
1994	\$26.89	\$15.50	2015	\$139.69	\$28.03
1995	\$30.60	\$16.13	2016	\$148.45	\$26.81
1996	\$33.10	\$17.13	2017	\$178.08	\$30.39
1997	\$36.71	\$18.00	2018	\$199.05	\$31.68
1998	\$44.90	\$17.38	2019	\$212.24	\$30.31
1999	\$44.74	\$16.06	2020	\$211.68	\$28.40
2000	\$47.98	\$17.31			

Table 2: Summary statistics of the main variables

This table reports summary statistics of the key variables used in the paper. *split_q1* is a dummy variable indicating whether the firm has announced a split in the next quarter. *split_yr* is a dummy variable indicating whether the firm has announced a split in the next four quarters. *ios* is the fraction of the firm's shares outstanding owned by institutional investors. *B/M* is the book to market ratio of equity. *atq* is the firm's size, as measured by total assets. *age* is the number of months since the firm first appears in CRSP. *prc* is the price level of the firm. *pct_split* is the percentage of firms in stock *i*'s industry that have conducted at least one split in the past year. *pplast* is the ratio of the current price to the price at last split. *Num_splits* is the number of times the stock has split in the past ten years. *LIQ* is firm's liquidity in the quarter, measured as the absolute daily return divided by the daily volume of trade (in millions), as in Amihud (2002). *VOL* is the standard deviation of firm *i*' daily returns in the quarter. *mom* is the cumulative returns of the stock in the past 12 months. *p_cme* is the log difference in the average market-to-book ratios of low- and high-priced stocks in the quarter, as in Baker, Greenwood, and Wurgler (2009).

Variables	N	mean	Std.Dev.	25 percentile	median	75 percentile
split_q1	375164	0.018181	0.133607	0	0	0
split_yr	375164	0.066342	0.248879	0	0	0
p_cme	375164	0.525595	0.208308	0.391672	0.519355	0.642735
pct_split	375164	0.074805	0.072177	0.015267	0.056047	0.116667
ios	375164	0.288251	0.373701	0	0	0.666146
Num_splits	375164	0.721962	1.186071	0	0	1
pplast	375164	2.27277	39.43033	0.680851	1.059406	1.652778
B/M	375164	0.5544	0.41747	0.267663	0.461922	0.741942
atq	375164	3183.533	11472.02	99.10475	361.9355	1601.292
prc	375164	66.69276	2650.228	10.375	18.79	34
age	375164	209.9883	197.6959	65	152	281
LIQ	375164	1.69E-07	5.52E-07	1.51E-09	1.11E-08	7.97E-08
VOL	375164	0.028351	0.014788	0.017886	0.024985	0.035112
mom	375164	0.244727	0.672797	-0.12794	0.117736	0.429035

Table 3: Predict stock splits in the next quarter

This table reports the performance of the following three prediction models in predicting splits in the next quarter: Gradient Boosting Machine (GBM), Random Forest (RF), and Logit Regressions (Logit). TP is the number of true positives, TN the number of true negatives, FP the number of false positives, and FN the number of false negatives. We allow for a 5% ex-ante false positive rate as follows: we first look at the predicted values among all nonsplitters in the training dataset (clean sample) and use the 95 percentile values as the cutoff values. We then look at the predicted values in the testing dataset and classify observations with predicted values above the cutoff as positives (i.e., splitters), and others as negatives (i.e., nonsplitters).

Model	TP	FP	FN	TN
Logit	147	6631	1160	172927
GBM	968	34752	339	144806
RF	562	11724	745	167834

Table 4: Portfolio returns from predicting stock splits in the next quarter

This table reports the performance of portfolios formed by the three prediction models in predicting splits in the next quarter. The Fama-French five-factor model is used to assess the abnormal returns. In Panels A and B, we form equal- and value-weighted portfolios, respectively. The *t*-statistics based on Newey-West standard errors with one lag are reported in parentheses under the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Performance of equal-weighted portfolios

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	-0.3042 (0.2054)	-0.0595 (0.1088)	-0.2351 (0.2243)
MKT-Rf	1.0318*** (0.0633)	0.9545*** (0.0360)	0.9689*** (0.0716)
SMB	0.3821*** (0.1027)	0.5757*** (0.0495)	0.7169*** (0.0974)
HML	-0.4072*** (0.1306)	-0.2335*** (0.0543)	-0.3583*** (0.1098)
RMW	-0.1911 (0.1589)	0.1405* (0.0721)	0.2510** (0.1268)
CMA	0.1097 (0.1960)	-0.1120 (0.1103)	-0.0830 (0.2168)
N	243	243	243
Adjusted R2	0.7379	0.9147	0.7410

Panel B: Performance of value-weighted portfolios

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	-0.1869 (0.1369)	-0.0277 (0.1128)	-0.0306 (0.2595)
MKT-Rf	0.9435*** (0.0468)	0.9693*** (0.0345)	0.9448*** (0.0933)
SMB	-0.3121*** (0.0497)	0.1144** (0.0536)	0.4242*** (0.1301)
HML	-0.3360*** (0.0724)	-0.2357*** (0.0628)	-0.4525*** (0.1498)
RMW	-0.0156 (0.1305)	0.2191** (0.0919)	0.3943** (0.1751)
CMA	0.0389 (0.1252)	-0.1959 (0.1224)	-0.2200 (0.2789)
N	243	243	243
Adjusted R2	0.8055	0.8702	0.5758

Table 5: Monthly returns around stock splits

In this table, we report the average monthly returns of all stock splits 11 months around the split announcement in the period of July 1, 2000 to December 31, 2020. Month 0 is the stock split announcement month, month -1 is the month before the announcement and so on.

<i>Month relative to split announcement</i>	<i>Return</i>
-11	4.21%
-10	4.95%
-9	4.85%
-8	4.24%
-7	4.32%
-6	5.25%
-5	5.00%
-4	5.35%
-3	6.01%
-2	5.66%
-1	5.75%
0	6.42%
1	1.92%
2	1.30%
3	0.66%
4	0.92%
5	1.52%
6	1.23%
7	1.15%
8	0.82%
9	1.03%
10	1.36%
11	0.77%

Table 6: Predict stock splits in the next quarter with SMOTE oversampling

This table reports the performance of the following three prediction models in predicting splits in the next quarter after oversampling the training data: Gradient Boosting Machine (GBM), Random Forest (RF), and Logit Regression (Logit). TP is the number of true positives, TN the number of true negatives, FP the number of false positives, and FN the number of false negatives. We allow for a 5% ex-ante false positive rate as follows: we first look at the predicted values among all nonsplitters in the training dataset (clean sample) and use the 95 percentile values as the cutoff values. We then look at the predicted values in the testing dataset and classify observations with predicted values above the cutoff as positives (i.e., splitters), and others as negatives (i.e., nonsplitters). Panel A reports the confusion matrixes of the predictions. Panel B reports the performance of portfolios formed by the three models. Panel C reports the performance of positive predictions of stocks in the smallest size quintile. The Fama-French five-factor model is used to assess the abnormal returns. The *t*-statistics based on Newey-West standard errors with one lag are reported in parentheses under the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Confusion matrixes from the prediction models with oversampling

Model	TP	FP	FN	TN
Logit	189	8719	1118	170839
GBM	1290	38182	17	141376
RF	269	2953	1038	176605

Panel B: Performance of portfolios of positive predictions

	Logit	GBM	RF	Logit	GBM	RF
	EW	EW	EW	VW	VW	VW
Intercept	-0.1329 (0.1564)	0.0973 (0.0596)	-0.2172 (0.3360)	-0.2242 (0.1664)	-0.0321 (0.0444)	-0.3506 (0.3913)
MKT-Rf	1.1671*** (0.0489)	1.0360*** (0.0199)	1.0440*** (0.1006)	1.0005*** (0.0510)	1.0028*** (0.0136)	1.0351*** (0.1110)
SMB	0.7090*** (0.0773)	0.6423*** (0.0268)	0.3855*** (0.1225)	-0.0973 (0.0912)	-0.0051 (0.024)	0.2039 (0.1422)
HML	-0.2707*** (0.0816)	-0.0582 (0.0362)	-0.1507 (0.2446)	-0.3185*** (0.0850)	-0.1737*** (0.0171)	-0.1024 (0.2631)
RMW	-0.3068*** (0.0774)	-0.0395 (0.0281)	0.3356** (0.1527)	0.0316 (0.1275)	0.1624*** (0.0257)	0.4583** (0.1880)
CMA	-0.2104* (0.1208)	-0.0643 (0.0406)	0.0008 (0.2208)	-0.1405 (0.1621)	0.0301 (0.0423)	-0.1895 (0.2584)
N	243	243	243	243	243	243
Adjusted R2	0.8968	0.9774	0.5302	0.7666	0.9779	0.4414

Panel C: Performance of positive predictions of stocks in the smallest size quintile

	Logit	GBM	RF	Logit	GBM	RF
	EW	EW	EW	VW	VW	VW
Intercept	0.0419 (0.5174)	0.2586** (0.1309)	0.9501** (0.4691)	0.1322 (0.5570)	0.0406 (0.1466)	0.8099* (0.4531)
MKT-Rf	1.0574*** (0.1538)	0.8707*** (0.0394)	0.9295*** (0.0847)	1.1078*** (0.1610)	0.8898*** (0.0517)	0.9231*** (0.0869)
SMB	0.9639*** (0.2218)	0.7527*** (0.0517)	0.4786*** (0.1824)	1.1027*** (0.2298)	0.8336*** (0.0681)	0.4768** (0.1935)
HML	-0.4243 (0.2684)	0.0587 (0.0695)	-0.3799* (0.1939)	-0.5877** (0.2681)	0.0761 (0.0677)	-0.3431** (0.1701)
RMW	0.1910 (0.3036)	0.0389 (0.0550)	0.0466 (0.1532)	0.2175 (0.2861)	0.1053 (0.0722)	0.0607 (0.1589)
CMA	-0.2495 (0.3086)	-0.0734 (0.0947)	-0.0451 (0.2352)	-0.1663 (0.3136)	-0.0472 (0.0995)	-0.1899 (0.2633)
N	243	243	243	243	243	243
Adjusted R2	0.3389	0.8734	0.3201	0.3499	0.8102	0.3617

Table 7: Predict stock splits in the next year

This table reports the performance of the following three prediction models in predicting splits in the next year: Gradient Boosting Machine (GBM), Random Forest (RF), and Logit Regression (Logit). TP is the number of true positives, TN the number of true negatives, FP the number of false positives, and FN the number of false negatives. We allow for a 5% ex-ante false positive rate as follows: we first look at the predicted values among all nonsplitters in the training dataset (clean sample) and use the 95 percentile values as the cutoff values. We then look at the predicted values in the testing dataset and classify observations with predicted values above the cutoff as positives (i.e., splitters), and others as negatives (i.e., nonsplitters).

Model	TP	FP	FN	TN
Logit	447	6881	4430	169107
GBM	4411	79570	466	96418
RF	1561	10246	3316	165742

Table 8: Portfolio returns from predicting splits in the next year

This table reports the performance of portfolios formed by the three prediction models in predicting splits in the next year. The Fama-French five-factor model is used to assess the abnormal returns. In Panels A and B, we form equal- and value-weighted portfolios, respectively. The *t*-statistics based on Newey-West standard errors with one lag are reported in parentheses under the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Performance of equal-weighted portfolios

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	-0.1497 (0.1267)	0.1396** (0.0576)	0.1237 (0.1436)
MKT-Rf	1.0639*** (0.0405)	1.0308*** (0.0200)	1.1090*** (0.0494)
SMB	0.3892*** (0.0694)	0.6171*** (0.0258)	0.6422*** (0.0716)
HML	-0.3135*** (0.0630)	-0.0864*** (0.0291)	-0.2784*** (0.0670)
RMW	-0.4105*** (0.0775)	-0.0020 (0.0302)	0.1128 (0.0781)
CMA	-0.0408 (0.1020)	-0.2047*** (0.0385)	-0.3164** (0.1376)
N	243	243	243
Adjusted R2	0.8899	0.9790	0.8902

Panel B: Performance of value-weighted portfolios

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	-0.1036 (0.1236)	0.0203 (0.0529)	-0.0055 (0.1841)
MKT-Rf	1.0054*** (0.0410)	1.0000*** (0.0121)	1.1214*** (0.0544)
SMB	-0.2816*** (0.0576)	0.0274 (0.0227)	0.2278** (0.0938)
HML	-0.3101*** (0.0659)	-0.2030*** (0.0232)	-0.4102*** (0.0888)
RMW	-0.1940** (0.0768)	0.0875*** (0.0265)	0.3689*** (0.1079)
CMA	-0.1738* (0.0918)	-0.1859*** (0.0358)	-0.4164*** (0.1609)
N	243	243	243
Adjusted R2	0.8784	0.9726	0.7984

Table 9: Portfolio returns from true positives and false positives

This table reports the performance of portfolios formed by the three prediction models in predicting splits in the next year, but we look at true positive predictions and false positive predictions separately. The Fama-French five-factor model is used to assess the abnormal returns. In Panels A and B, we form equal- and value-weighted portfolios of true positive predictions, respectively. In Panel C, we form value-weighted portfolios of true positive predictions from Logit with three different models: CAPM, 3-factor, and 5-factor models. In Panels D and E, we form equal- and value-weighted portfolios of false positive predictions, respectively. The *t*-statistics based on Newey-West standard errors with one lag are reported in parentheses under the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Performance of equal-weighted true positive portfolios

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	1.0401*** (0.3566)	1.6705*** (0.1908)	1.2663*** (0.2642)
MKT-Rf	1.2741*** (0.1273)	1.0205*** (0.0661)	1.1112*** (0.0967)
SMB	0.7208*** (0.1513)	0.5650*** (0.0677)	0.5765*** (0.1077)
HML	-0.7168*** (0.2082)	-0.2966*** (0.1018)	-0.4455*** (0.1242)
RMW	0.5763** (0.2274)	0.4183*** (0.0934)	0.4822*** (0.1416)
CMA	0.1845 (0.2677)	0.0094 (0.1273)	0.0138 (0.1694)
N	243	243	243
Adjusted R2	0.5493	0.7815	0.6678

Panel B: Performance of value-weighted true positive portfolios

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	0.4747 (0.3567)	0.9691*** (0.2258)	0.8891*** (0.2910)
MKT-Rf	1.2339*** (0.1001)	1.1813*** (0.0912)	1.1733*** (0.1135)
SMB	0.2033 (0.1698)	0.0922 (0.0851)	0.2169* (0.1115)
HML	-0.8935*** (0.1598)	-0.5200*** (0.1224)	-0.6596*** (0.1251)
RMW	0.6359*** (0.1962)	0.6641*** (0.1489)	0.5264*** (0.1630)
CMA	0.5094** (0.2497)	0.2865* (0.1692)	0.2942* (0.1732)
N	243	243	243
Adjusted R2	0.4703	0.6761	0.5878

Panel C: Performance of value-weighted portfolios of TP Logit predictions

	Logit	Logit	Logit
	(1) CAMP	(2) 3-Factor	(3) 5-Factor
Intercept	0.9798*** (0.3657)	0.9666*** (0.3415)	0.4747 (0.3567)
MKT-Rf	1.0435*** (0.0994)	1.0447*** (0.0945)	1.2339*** (0.1001)
SMB		0.0756 (0.1761)	0.2033 (0.1698)
HML		-0.5243*** (0.1645)	-0.8935*** (0.1598)
RMW			0.6359*** (0.1962)
CMA			0.5094** (0.2497)
N	243	243	243
Adjusted R2	0.3961	0.4345	0.4703

Panel D: Performance of equal-weighted false positive portfolios

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	-0.2578** (0.1225)	0.0278 (0.0552)	-0.1178 (0.1400)
MKT-Rf	1.0631*** (0.0397)	1.0333*** (0.0194)	1.1160*** (0.0484)
SMB	0.3703*** (0.0666)	0.6087*** (0.0266)	0.6157*** (0.0731)
HML	-0.3110*** (0.0629)	-0.0889*** (0.0277)	-0.2819*** (0.0659)
RMW	-0.4579*** (0.0756)	-0.0345 (0.0305)	0.0480 (0.0797)
CMA	-0.0490 (0.1002)	-0.2223*** (0.0372)	-0.3529** (0.1398)
N	243	243	243
Adjusted R2	0.8984	0.9803	0.8949

Panel E: Performance of value-weighted false positive portfolios

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	-0.1276 (0.1208)	-0.0596 (0.0512)	-0.2027 (0.1881)
MKT-Rf	0.9800*** (0.0388)	0.9854*** (0.0131)	1.0783*** (0.0551)
SMB	-0.2802*** (0.0576)	0.0204 (0.0234)	0.2167** (0.0991)
HML	-0.2829*** (0.0656)	-0.1927*** (0.0257)	-0.3582*** (0.0935)
RMW	-0.2330*** (0.0705)	0.0249 (0.0303)	0.3265*** (0.1155)
CMA	-0.2024** (0.0911)	-0.2320*** (0.0408)	-0.5157*** (0.1682)
N	243	243	243
Adjusted R2	0.8786	0.9735	0.7838

Table 10: Performance of positive prediction portfolios among small firms

This table reports the performance of portfolios formed by the three prediction models in predicting splits in the next year, but we focus on stocks in the smallest size quintile. The Fama-French five-factor model is used to assess the abnormal returns. In Panels A and B, we form equal- and value-weighted portfolios of positive predictions in the smallest size quintile, respectively. The *t*-statistics based on Newey-West standard errors with one lag are reported in parentheses under the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Performance of equal-weighted portfolios of positive predictions of small stocks

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	0.5481 (0.7089)	0.3765*** (0.1172)	-0.4929 (0.3953)
MKT-Rf	1.0983*** (0.2250)	0.8045*** (0.0375)	1.0875*** (0.1086)
SMB	0.9441*** (0.2121)	0.7237*** (0.0448)	0.4334** (0.1831)
HML	0.3162 (0.3932)	0.0470 (0.0577)	-0.0866 (0.1773)
RMW	-0.6980** (0.3133)	0.1089** (0.0500)	0.1910 (0.2094)
CMA	-1.0370* (0.5303)	-0.1401* (0.0810)	0.5088* (0.2903)
N	243	243	243
Adjusted R2	0.3931	0.8755	0.4390

Panel B: Performance of value-weighted portfolios of positive predictions of small stocks

	Logit	GBM	RF
	(1)	(2)	(3)
Intercept	0.6630 (0.7875)	0.3305** (0.1289)	-0.5723 (0.4208)
MKT-Rf	1.0968*** (0.2624)	0.8464*** (0.0385)	1.0619*** (0.1275)
SMB	0.8480*** (0.2252)	0.7775*** (0.0493)	0.4480** (0.2091)
HML	0.2862 (0.4093)	0.0587 (0.0596)	0.0367 (0.2239)
RMW	-0.5847 (0.3655)	0.1601*** (0.0586)	0.2731 (0.2264)
CMA	-1.1873** (0.5604)	-0.1104 (0.0774)	0.3502 (0.3212)
N	243	243	243
Adjusted R2	0.3390	0.8317	0.4112

Table 11: Portfolio abnormal returns using other factor models

This table reports the performance of portfolios formed by GBM predictions of splits in the next year. Four different factor models are used to assess abnormal returns: CAPM, Fama French three-factor model, Fama French five-factor model, and Fama French five-factor model augmented by the momentum factor. In Panels A and B, we form equal- and value-weighted portfolios, respectively. In Panels C and D, we form equal- and value-weighted portfolios of stocks in the smallest size quintile, respectively. The *t*-statistics based on Newey-West standard errors with one lag are reported in parentheses under the coefficients. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Panel A: Performance of equal-weighted portfolios

	CAPM	FF 3-Factor	FF 5-Factor	6-Factor
	(1)	(2)	(3)	(4)
Intercept	0.1870*	0.0901	0.1396**	0.1397**
	(0.1077)	(0.0573)	(0.0576)	(0.0573)
MKT-Rf	1.1718***	1.0510***	1.0308***	1.0230***
	(0.0312)	(0.0212)	(0.0200)	(0.0199)
SMB		0.6188***	0.6171***	0.6175***
		(0.0265)	(0.0258)	(0.0254)
HML		-0.1670***	-0.0864***	-0.0969***
		(0.0264)	(0.0291)	(0.0300)
RMW			-0.0020	0.0111
			(0.0302)	(0.0304)
CMA			-0.2047***	-0.1960***
			(0.0385)	(0.0379)
MOM				-0.0237
				(0.0175)
N	243	243	243	243
Adjusted R2	0.9041	0.9760	0.9790	0.9793

Panel B: Performance of value-weighted portfolios

	CAPM	FF 3-Factor	FF 5-Factor	6-Factor
	(1)	(2)	(3)	(4)
Intercept	0.0300	0.0275	0.0203	0.0201
	(0.0790)	(0.0514)	(0.0529)	(0.0509)
MKT-Rf	0.9930***	0.9987***	1.0000***	1.0151***
	(0.0237)	(0.0146)	(0.0121)	(0.0107)
SMB		0.0115	0.0274	0.0265
		(0.0231)	(0.0227)	(0.0216)
HML		-0.2524***	-0.2030***	-0.1825***
		(0.0173)	(0.0232)	(0.0236)
RMW			0.0875***	0.0619***
			(0.0265)	(0.0227)
CMA			-0.1859***	-0.2030***
			(0.0358)	(0.0335)
MOM				0.0463***
				(0.0121)
N	243	243	243	243
Adjusted R2	0.9414	0.9679	0.9726	0.9744

Panel C: Performance of equal-weighted portfolios of small stocks

	CAPM	FF 3-Factor	FF 5-Factor	6-Factor
	(1)	(2)	(3)	(4)
Intercept	0.5165*** (0.1714)	0.4070*** (0.1158)	0.3765*** (0.1172)	0.3765*** (0.1171)
MKT-Rf	0.9381*** (0.0494)	0.7942*** (0.0352)	0.8045*** (0.0375)	0.7979*** (0.0392)
SMB		0.7034*** (0.0459)	0.7237*** (0.0448)	0.7241*** (0.0446)
HML		0.0213 (0.0505)	0.0470 (0.0577)	0.0380 (0.0602)
RMW			0.1089** (0.0500)	0.1201** (0.0516)
CMA			-0.1401* (0.0810)	-0.1326* (0.0777)
MOM				-0.0203 (0.0290)
N	243	243	243	243
Adjusted R2	0.7443	0.8729	0.8755	0.8753

Panel D: Performance of value-weighted portfolios of small stocks

	CAPM	FF 3-Factor	FF 5-Factor	6-Factor
	(1)	(2)	(3)	(4)
Intercept	0.5142*** (0.1882)	0.3980*** (0.1306)	0.3305** (0.1289)	0.3303** (0.1283)
MKT-Rf	0.9759*** (0.0553)	0.8219*** (0.0381)	0.8464*** (0.0385)	0.8617*** (0.0375)
SMB		0.7469*** (0.0479)	0.7775*** (0.0493)	0.7766*** (0.0491)
HML		0.0583 (0.0551)	0.0587 (0.0596)	0.0794 (0.0587)
RMW			0.1601*** (0.0586)	0.1343** (0.0610)
CMA			-0.1104 (0.0774)	-0.1277* (0.0753)
MOM				0.0467 (0.0346)
N	0.6987	0.8287	0.8317	0.8325
Adjusted R2	243	243	243	243